

Dimitrios – Georgios **Kontopoulos**

WALTER BENJAMIN POSTDOCTORAL FELLOW (HE/HIM)

Department of Ecology and Evolutionary Biology, University of California, Los Angeles, CA, USA

+1 424-402-7790 | dgkontopoulos@gmail.com | dgkontopoulos.io | [@DGKontopoulos@ecoevo.social](https://twitter.com/DGKontopoulos) |  Dimitrios – Georgios Kontopoulos

I am an integrative biologist. My research focuses on understanding **how diverse biological systems (from molecules to ecosystems) respond to environmental changes over various timescales**. I approach this goal using approaches from a wide range of fields, including ecoinformatics, phylogenetic comparative methods, comparative genomics, and behavioral experiments.

Research appointments and internships

- **Walter Benjamin Postdoctoral Fellow at Prof. Noa Pinter-Wollman's group**, University of California, Los Angeles, USA. **June 2025 - Present**
- **Postdoctoral researcher at Prof. Michael Hiller's group**, LOEWE Centre for Translational Biodiversity Genomics & Senckenberg Research Institute, Frankfurt am Main, Germany. **May 2021 - Dec. 2024**
- **Visiting researcher**, Imperial College London, Silwood Park, Ascot, United Kingdom. **Dec. 2019 - Apr. 2021**
- **Research assistant at Prof. Samraat Pawar's group**, Imperial College London, Silwood Park, Ascot, United Kingdom. **Oct. 2015 - Sep. 2016**
Nov. 2014 - May 2015
- **Postgraduate intern at Dr. Sofia Kossida's group**, Bioinformatics and Medical Informatics Lab of the Biomedical Research Foundation of the Academy of Athens, Athens, Greece. **Nov. 2012 - Sep. 2013**
- **Summer intern at Prof. Marie-Paule Lefranc's group**, Laboratoire d'ImmunoGénétique Moléculaire of the Institut de Génétique Humaine, Montpellier, France. **May - June 2013**
- **Summer intern at Prof. Zissis Mamuris' group**, Laboratory of Genetics, Comparative and Evolutionary Biology of the Department of Biochemistry and Biotechnology of the University of Thessaly, Larissa, Greece. **July 2011**
- **Intern at Dr. George Skavdis' group**, Laboratory of Molecular Regulation of the Department of Molecular Biology and Genetics of the Democritus University of Thrace, Alexandroupolis, Greece. **Mar. - May 2010**

Education

- **Imperial College London**, Silwood Park, Ascot, United Kingdom **Oct. 2015 - Dec. 2019**
PhD: "Limits to thermal adaptation in ectotherms"
- **Imperial College London**, Silwood Park, Ascot, United Kingdom **Sep. 2013 - Sep. 2014**
MRes Biodiversity Informatics and Genomics, graduated with Distinction.
Thesis: "Phylogenetic constraints and environmental drivers of thermal adaptation among the phytoplankton"
- **Democritus University of Thrace**, Alexandroupolis, Greece **Sep. 2008 - Oct. 2012**
BSc Molecular Biology and Genetics, graduated with 7.46/10 ("Very Well").
Thesis: "Pinda: a gene duplication detection program"

Publications

Peer-reviewed papers († equal contribution; ✉ corresponding author)

- 19 Yi, X. ✉, **Kontopoulos, D.-G.**, & Hiller, M. ✉ (2025) [Comprehensive phylogenetic trait estimations support ancestral omnivory in the ecologically diverse bat family Phyllostomidae](#). *Evolution*. 79(11):2406-2420.
- 18 **Kontopoulos, D.-G.** ✉, Patmanidis, I., Barraclough, T.G., & Pawar, S. (2025) [Changes in flexibility but not in compactness underlie the thermal adaptation of prokaryotic adenylate kinases](#). *Evolution Letters*. 9(5):598-609.
- 17 Morales, A.E.†, Dong, Y.†, Brown, T., Baid, K., **Kontopoulos, D.-G.**, Gonzalez, V., Huang, Z., Ahmed, A.-W., Bhuinya, A., Hilgers, L.,

- Winkler, S., Hughes, G., Li, X., Lu, P., Yang, Y., Kirilenko, B.M., Devanna, P., Lama, T.M., Nissan, Y., Pippel, M., Dávalos, L.M., Vernes, S.C., Puechmaille, S.J., Rossiter, S.J., Yovel, Y., Prescott, J.B., Kurth, A., Ray, D.A., Lim, B.K., Myers, E., Teeling, E.C., Banerjee, A., Irving, A.T. , & Hiller, M.  (2025) [Bat genomes illuminate adaptations to viral tolerance and disease resistance](#). *Nature*. 638:449-458.
- 16 **Kontopoulos, D.-G.** , Levesque, D.L., & Hiller, M.  (2025) [Numerous independent gains of daily torpor and hibernation across endotherms, linked with adaptation to diverse environments](#). *Functional Ecology*. 39(3):824-839.
 - 15 **Kontopoulos, D.-G.** , Sentis, A., Daufresne, M., Glazman, N., Dell, A.I., & Pawar, S. (2024) [No universal mathematical model for thermal performance curves across traits and taxonomic groups](#). *Nature Communications*. 15:8855. **[Co-second place winner of the 2025 Outstanding Paper Award from the Early Career Ecologists Section of the Ecological Society of America]**
 - 14 Pawar, S. , Huxley, P.J. , Smallwood, T.R.C., Nesbit, M.L., Chan, A.H.H., Shocket, M.S., Johnson, L.R., **Kontopoulos, D.-G.**, & Cator, L.J.  (2024) [Variation in temperature of peak trait performance constrains adaptation of arthropod populations to climatic warming](#). *Nature Ecology & Evolution*. 8:500-510.
 - 13 Kirilenko, B.M., Munegowda, C., Osipova, E., Jebb, D., Sharma, V., Blumer, M., Morales, A.E., Ahmed, A.-W., **Kontopoulos, D.-G.**, Hilgers, L., Lindblad-Toh, K., Karlsson, E.K., Zoonomia Consortium, & Hiller, M.  (2023) [Integrating gene annotation with orthology inference at scale](#). *Science*. 380(6643):eabn3107.
 - 12 Smith, T.P. , Mombrikotb, S., Ransome, E., **Kontopoulos, D.-G.**, Pawar, S., & Bell, T. (2022) [Latent functional diversity may accelerate microbial community responses to temperature fluctuations](#). *eLife*. 11:e80867.
 - 11 Kordas, R.L., Pawar, S., **Kontopoulos, D.-G.**, Woodward, G., & O’Gorman, E.J.  (2022) [Metabolic plasticity can amplify ecosystem responses to global warming](#). *Nature Communications*. 13:2161.
 - 10 **Kontopoulos, D.-G.** , Smith, T.P., Barraclough, T.G., & Pawar, S. (2020) [Adaptive evolution shapes the present-day distribution of the thermal sensitivity of population growth rate](#). *PLOS Biology*. 18(10):e3000894.
 - 9 **Kontopoulos, D.-G.** , van Sebillie, E., Lange, M., Yvon-Durocher, G., Barraclough, T.G., & Pawar, S. (2020) [Phytoplankton thermal responses adapt in the absence of hard thermodynamic constraints](#). *Evolution*. 74(4):775-790. **[Top cited article 2020-2021 in Evolution]**
 - 8 García-Carreras, B. , Sal, S., Padfield, D., **Kontopoulos, D.-G.**, Bestion, E., Schaum, C.-E., Yvon-Durocher, G., & Pawar, S.  (2018) [Role of carbon allocation efficiency in the temperature dependence of autotroph growth rates](#). *Proceedings of the National Academy of Sciences*. 115(31):E7361-E7368.
 - 7 Kumbhar, R., Vidal-Eychenié, S., **Kontopoulos, D.-G.**, Larroque, M., Larroque, C., Basbous, J., Kossida, S., Ribeyre, C., & Constantinou, A.  (2018) [Recruitment of ubiquitin-activating enzyme UBA1 to DNA by poly\(ADP-ribose\) promotes ATR signalling](#). *Life Science Alliance*. 1(3):e201800096.
 - 6 **Kontopoulos, D.-G.** , García-Carreras, B., Sal, S., Smith, T.P., & Pawar, S. (2018) [Use and misuse of temperature normalisation in meta-analyses of thermal responses of biological traits](#). *PeerJ*. 6:e4363.
 - 5 **Kontopoulos, D.-G.** , Kontopoulou, T., Ho, H.-C., & García-Carreras, B. (2017) [Towards a theoretically informed policy against a rakghoul plague outbreak](#). *The Medical Journal of Australia*. 207(11):490-494. **[Third place in the 2017 Christmas Competition of the Medical Journal of Australia]**
 - 4 **Kontopoulos, D.-G.** , Vlachakis, D. , Tsiliki, G., & Kossida, S. (2016) [Structuprint: a scalable and extensible tool for two-dimensional representation of protein surfaces](#). *BMC Structural Biology*. 16:4.
 - 3 Kontopoulou, T. , **Kontopoulos, D.-G.** , Vaidakis, E., & Mousoulis, G.P. (2015) [Adult Kawasaki disease in a European patient: a case report and review of the literature](#). *Journal of Medical Case Reports*. 9(1):75.
 - 2 Vlachakis, D., **Kontopoulos, D.-G.**, & Kossida, S.  (2013) [Space Constrained Homology Modelling: the paradigm of the RNA-dependent RNA polymerase of dengue \(type II\) virus](#). *Computational and Mathematical Methods in Medicine*. 2013:108910.
 - 1 **Kontopoulos, D.-G.** & Glykos, N.M.  (2013) [Pinda: a web service for detection and analysis of intraspecies gene duplication events](#). *Computer Methods and Programs in Biomedicine*. 111(3):711-714.

Commentaries

- 1 Pawar, S. ✉ & **Kontopoulos, D.-G.** (2025) [Toward a general understanding of thermal performance curves in biology](#). *Proceedings of the National Academy of Sciences*. 122(51):e2528528122.

Invited book chapters

- 1 **Kontopoulos, D.-G.** ✉ Phylogenetic comparative approaches for the study of biological scaling. In: Synthesizing biological scaling: towards a universal theory. Santa Fe Institute Press. *In press*.

Manuscripts under review

- 2 Hilgers, L. ✉, Rovatsos, M., **Kontopoulos, D.-G.**, Brown, T., Hickler, T., Huntley, B., Pippel, M., Munegowda, C., Müller, T., Ahmed, A.-W., Laas, A., Praschag, P., Damas, J., Winkler, S., Lewin, H., Myers, E., Fritz, U., & Hiller, M. ✉ Side-necked turtle genomes reveal chromosomal dynamics, skeletal innovation and cancer resistance. Preprint available from bioRxiv: [doi:10.64898/2026.03.05.709825](#).
- 1 **Kontopoulos, D.-G.** ✉, Ahmed, A.-W., Bein, B., Levesque, D.L., & Hiller, M. ✉ Comparative genomics indicate multiple genetic routes to the evolution of torpor in placental mammals. Preprint available from bioRxiv: [doi:10.1101/2025.11.18.689136](#).

Fellowships, scholarships, and awards

Total funds secured: >€380,000

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| 9 Scholarship Award from the Hellenic Bioscientific Association of the USA for attending the 2026 Annual Meeting of the Ecological Society of America. \$500 | Apr. 2026 |
| 8 Outstanding Paper Award (co-second place winner) from the Early Career Ecologists Section of the Ecological Society of America for peer-reviewed paper #15 above. | Aug. 2025 |
| 7 Walter Benjamin Postdoctoral Fellowship from the German Research Foundation (DFG) . €133,585 | June 2025 - May 2027 |
| 6 EMBO Postdoctoral Fellowship. €168,000 | Mar. 2022 - Apr. 2024 |
| 5 Top cited article award from Evolution for peer-reviewed paper #9 above. | Mar. 2022 |
| 4 Third place in the Christmas Competition of the Medical Journal of Australia for peer-reviewed paper #5 above. | Dec. 2017 |
| 3 Travel award from the Department of Life Sciences, Imperial College London for attending the 2017 Congress of the European Society for Evolutionary Biology in Groningen, the Netherlands. £500 | May 2017 |
| 2 Science and Solutions for a Changing Planet Doctoral Training Partnership scholarship from the Natural Environment Research Council . £57,300 | Oct. 2015 - Apr. 2019 |
| 1 Scholarship for 2013-2014 postgraduate education abroad (1st cycle) from the Greek State Scholarships Foundation (IKY) . €13,680 | Dec. 2013 |

Presentations

Invited talks

- 4 **Evolution of ecophysiological responses to temperature changes.** Université Clermont Auvergne, France, 10th November 2023.
- 3 **Deep-time evolution of biological responses to temperature changes.** Ecology & Evolution Seminar Series, Imperial College London, Silwood Park Campus, United Kingdom, 10th October 2019.
- 2 **Deep-time evolution of physiological responses to temperature changes.** Stanford University, CA, United States of America, 13th September 2019.
- 1 **Trait correlations vs environmental drivers in the evolution of phytoplankton thermal responses.** National Taiwan University, Taiwan, 26th March 2018.

Contributed talks

- 8 **Kontopoulos, D.-G.**, Ahmed, A.-W., Bein, B., & Hiller, M. (2024) The impact of phylogenetic diversity on the strength of gene-trait associations. *18th Conference of the Hellenic Society for Computational Biology and Bioinformatics, Lamia, Greece, 17-19th October.*
- 7 **Kontopoulos, D.-G.**, Levesque, D.L., & Hiller, M. (2023) Physiological, ecological, and genomic underpinnings of daily torpor and hibernation across mammals and birds. *2023 Annual Meeting of the Ecological Society of America, Portland, OR, United States of America, 6th-11th August.*
- 6 **Kontopoulos, D.-G.**, van Sebillie, E., Lange, M., Yvon-Durocher, G., Barraclough, T.G., & Pawar, S. (2018) Non-random adaptive evolution of the thermal sensitivity of growth rate among phytoplankton. *Gordon Research Seminar on Unifying Ecology Across Scales, Biddeford, ME, United States of America, 21st-22nd July.*
- 5 **Kontopoulos, D.-G.**, van Sebillie, E., Lange, M., Yvon-Durocher, G., & Pawar, S. (2018) Trait correlations vs environmental drivers in the evolution of phytoplankton thermal responses. *65th Annual Meeting of the Ecological Society of Japan, Sapporo, Japan, 14th-18th March.*
- 4 **Kontopoulos, D.-G.**, Yvon-Durocher G., & Pawar, S. (2017) Niche convergence in the macroevolution of the thermal sensitivity of phytoplankton growth rate. *2017 Congress of the European Society for Evolutionary Biology, Groningen, the Netherlands, 20th-25th August.*
- 3 **Kontopoulos, D.-G.**, Yvon-Durocher, G., & Pawar, S. (2016) Deep-time macroevolution of thermal sensitivity of growth rate among phytoplankton. *Annual Meeting of the British Ecological Society, Liverpool, United Kingdom, 11th-14th December.*
- 2 **Kontopoulos, D.-G.**, Yvon-Durocher, G., Chen, B., Thomas, M. K. & Pawar S. (2014) Γενικά μοτίβα θερμικής προσαρμογής μεταξύ των ειδών του φυτοπλαγκτού [General patterns of thermal adaptation among phytoplankton]. *7th National Congress of the Hellenic Ecological Society, Mytilene, Greece, 9th-12th October.*
- 1 **Kontopoulos, D.-G.** & Glykos, N.M. (2012) Pinda: a web service for detection and analysis of intraspecies gene duplications. *7th Conference of the Hellenic Society for Computational Biology and Bioinformatics, Heraklion, Greece, 4th-6th October.*

Contributed posters

- 5 **Kontopoulos, D.-G.**, Tran, C., Barreras, S., & Pinter-Wollman, N. (2026) Effects of temperature and relative humidity on the behavior of Argentine ants. *15th Annual Southern California Systems Biology Symposium, Los Angeles, CA, United States of America, 25th April.*
- 4 **Kontopoulos, D.-G.**, Tran, C., Barreras, S., & Pinter-Wollman, N. (2026) Effects of temperature and relative humidity on the behavior of Argentine ants. *Southern California Animal Behavior Meeting, Claremont, CA, United States of America, 18th April.*
- 3 **Kontopoulos, D.-G.**, Patmanidis, I., Barraclough, T.G., & Pawar, S. (2018) Nonsynonymous mutations are more detrimental at high temperatures; a prokaryote-wide study of adenylate kinases. *Gordon Research Conference on Unifying Ecology Across Scales, Biddeford, ME, United States of America, 22nd-27th July.*
- 2 **Kontopoulos, D.-G.**, Yvon-Durocher, G., & Pawar, S. (2016) Deep-time macroevolution of thermal sensitivity of growth rate among phytoplankton. *Gordon Research Conference on Unifying Ecology Across Scales, Biddeford, ME, United States of America, 24th-29th July.*
- 1 **Kontopoulos, D.-G.**, Yvon-Durocher, G., Allen, A.P., Chen, B., Thomas, M.K., & Pawar, S. (2014) Phylogenetic constraints and environmental drivers of thermal adaptation among the phytoplankton. *Annual London Evolutionary Research Network Conference, London, United Kingdom, 5th November.*

Research skills

Global change biology

- quantifying the shape of thermal performance curves for physiological, ecological, or other biological traits through diverse (80+) nonlinear mathematical models.
- identifying trait-trait and trait-environment associations through phylogenetic generalised linear mixed models with numerous (up to 22 so far) covarying response variables.
- sampling of *Linepithema humile* ants from natural colonies, conducting behavioral experiments at multiple temperatures and humidity levels, and analysing the data using Artificial Intelligence software.

Statistics & data science

- Bayesian statistics.
- phylogenetic comparative methods.
- likelihood-based model selection.
- dimensionality reduction and clustering.
- machine learning.

Bioinformatics

- genome alignment and annotation.
- Gene Ontology term enrichment.
- genome-wide screening for signatures of selection, gene losses, or evolutionary rate shifts, associated with a trait of interest.
- analysis of sequence or physicochemical conservation.
- protein structure modelling and comparison.
- molecular dynamics simulations.

Phylogenetics

- likelihood-based reconstruction of gene and species trees.
- timetree inference.

Ecological modelling

- modelling predator-prey population dynamics using ordinary differential equations.
- agent-based modelling with NetLogo.

Scientific programming

- extensive experience in Perl and R.
- good experience in Python and SQL.
- basic experience in Common Lisp, C, and Shell.
- version control using Git.
- some experience in web development.

Teaching

Certification

- **Associate Teaching Certificate**, Center for the Integration of Research, Teaching, and Learning (CIRTL) **2026**

Experience as a course demonstrator

- **Further Topics in Statistics** **2015-18**
MSc/MRes “Ecology, Evolution and Conservation”, Imperial College London
- **Intro to UNIX and Linux** **2017**
MSc/MRes “Computational Methods in Ecology and Evolution” and “Quantitative and Modelling Skills in Ecology and Evolution” Centre for Doctoral Training, Imperial College London
- **Statistics** **2014-15**
BSc “Biological Sciences”, year 1, Imperial College London
- **Biological Computing in Python II** **2014**
MSc/MRes “Computational Methods in Ecology and Evolution”, Imperial College London
- **Computational Biostatistics** **2014**
BSc “Biological Sciences”, year 2, Imperial College London

Experience as a course tutor

- **MSc/MRes “Computational Methods in Ecology and Evolution”**, Imperial College London **2014-15**

Experience as a workshop presenter

- **“How to generate topological constraints using the Open Tree of Life”** **30th Mar. 2017**
Silwood Computer Skillz Workshop, Imperial College London

Mentoring

Undergraduate interns

- **Sofia Barreras** - BS "Atmospheric and Oceanic Sciences", University of California, Los Angeles. **2026**
Duties: determining ant counts and body lengths from images using ImageJ.
- **Catherine Tran** - BS "Ecology, Evolution, and Behavior", University of California, Los Angeles. **2025-26**
Duties: determining ant counts from images using ImageJ.

Master's students

- **Georgios Kalogiannis** - MRes "Computational Methods in Ecology and Evolution", Imperial College London. **2024**
Thesis: "Gene loss is an important signature of insect size evolution"
Primary supervisor: Samraat Pawar
- **Aditi Madkaikar** - MRes "Computational Methods in Ecology and Evolution", Imperial College London. **2023**
Thesis: "Predicting the thermal niche of a ubiquitous bacterium using whole genome sequence"
Primary supervisor: Samraat Pawar
Other supervisors: Arianna Basile
- **Kate Griffin** - MSc "Computational Methods in Ecology and Evolution", Imperial College London. **2022**
Thesis: "Can't stand the heat? An analysis of the thermal sensitivity of arthropods, how it has evolved & factors influencing it"
Primary supervisor: Samraat Pawar
Other supervisors: Paul Huxley, Lauren Cator

Selected media coverage

Mentions in popular press

- [Feature](#) in the "Feedback" column of the **New Scientist** for peer-reviewed paper #16 above. **5th Feb. 2025**

Institutional press releases and highlights

- [Press release](#) from the **Senckenberg Society for Nature Research** for peer-reviewed paper #16 above. **27th Jan. 2025**
Distributed by the [idw](#) ("Science Information Service") and the [VBIO](#) ("German Life Sciences Association").
- [Highlight](#) from **Imperial College London** for peer-reviewed paper #10 above. **13th Nov. 2020**

Outreach / public engagement

- Exhibitor at the **Great Exhibition Road Festival**, London, UK **7th-8th May 2016**
- Co-organiser of the "Drawing Climate Change" activity at the **Science Museum Lates**, London, UK **30th Mar. 2016**

Service

Manuscript reviewer for 16 Journals / peer review platforms: *Communications Biology*, *Communications Earth & Environment*, *Ecology Letters*, *eLife*, *Frontiers in Microbiology*, *Functional Ecology*, *Global Ecology and Biogeography*, *Journal of Plankton Research*, *Journal of Thermal Biology*, *Life Science Alliance*, *Limnology and Oceanography*, *Physiological and Biochemical Zoology*, *Proceedings of the Royal Society B: Biological Sciences*, *Review Commons*, *Scientific Reports*, and *Systematic Biology*.

Scientific workshops and courses attended

- 8 Decentering Grades: Getting Started with Ungrading for Future Faculty**, CIRTl Net-work, held online. **2nd Mar. 2026**
- 7 Maximizing Learning Outcomes from Instructional Demonstration: Balancing Instructor- and Student-Led Experiential Learning**, CIRTl Network, held online. **17th Feb. 2026**
- 6 CIRTl@UCLA Introduction to Evidence-Based Undergraduate Teaching course**, University of California, Los Angeles, CA, USA. **Sep. 2025 - Dec. 2025**

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| 5 CIMER/NRMN Postdoc Research Mentoring Workshop , University of California, Los Angeles, CA, USA. | 28th Aug. 2025 |
| 4 An Introduction to Mechanistic Niche Modelling with NicheMapR and TrenchR , 2023 Annual Meeting of the Ecological Society of America, Portland, OR, United States of America. | 6th Aug. 2023 |
| 3 EMBO Laboratory Leadership course , EMBO Solutions, Heidelberg, Germany. | 5th-7th July 2023 |
| 2 “Introduction to Agent-Based Modelling” Massive Open Online Course , Santa Fe Institute, held online. | June 2022 - Aug. 2022 |
| 1 Evolutionary Quantitative Genetics workshop , Friday Harbor Laboratories, University of Washington, held online. | 11th-15th July 2022 |

Language skills

Native proficiency in **Greek**, full proficiency in **English**, sufficient proficiency in **French**, basic proficiency in **German**.

References

Prof. Samraat Pawar

Title: Professor of Theoretical Ecology

Affiliation: Department of Life Sciences, Imperial College London, Silwood Park

Email address: s.pawar@imperial.ac.uk

Prof. Michael Hiller

Title: Professor of Comparative Genomics

Affiliation: Senckenberg Research Institute, Frankfurt and Goethe University

Email address: michael.hiller@senckenberg.de

Prof. Noa Pinter-Wollman

Title: Professor

Affiliation: Department of Ecology and Evolutionary Biology, University of California, Los Angeles

Email address: nmpinter@ucla.edu

Prof. Timothy G. Barraclough

Title: Professor of Evolutionary Biology

Affiliation: Department of Biology, University of Oxford

Email address: tim.barraclough@biology.ox.ac.uk

Additional information

Member of Scientific Societies: Society for the Study of Evolution, Ecological Society of America, Panhellenic Association of Bioscientists, Hellenic Bioscientific Association of the USA.

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